

# One-step lookahead in safety-gated dual-agent dose-finding: an operating-characteristic study and a decision rule

Xinzhu Wang | Tanzy Love  
xinzhu\_wang@urmc.rochester.edu

Department of Biostatistics and Computational Biology, University of Rochester

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MAIN RESULT: one-step lookahead buys recommended-dose efficacy - not a universal improvement in localization or safety.

## 1. Clinical question / why pharma should care

Early-phase combination trials must identify an optimal biological dose (OBD): the most efficacious dose pair whose toxicity remains acceptable.

- Recommended-dose efficacy, OBD localization, final recommendation safety, and in-trial exposure are distinct operating characteristics.
- We hold the surrogate, gate, grid, budget, and recommendation rule fixed; only the next-cohort acquisition changes.
- Project Optimus (oncology): optimize dosage rather than default to the maximum tolerated dose.

## 2. Safety-gated dose-finding setup

$$d^*(z) = \arg \max_{d \in [0,1]^2} e(d, z) \quad \text{s.t.} \quad g(d, z) \leq g^\dagger(z)$$

$$\text{PF}(d) = \Phi\left(\frac{g^\dagger - \mu_g(d)}{\sigma_g(d)}\right), \quad A_\tau = \{d : \text{PF}(d) > \tau\}$$

$$d_{\text{next}} = \arg \max_{d \in A_\tau \cap G} \alpha(d), \quad \hat{d} = \arg \max_{d \in A_\tau} \mu_e(d)$$

### Benchmark design

2 agents; 2 severity strata    5 × 5 grid    N = 40 / stratum    cohort = 2    4 warm-up

Displayed gate levels:  $\tau = 0.5, 0.7, 0.9$  | Empty gate: max-PF fallback | No early stopping | n: OSA 200/cell; logistic/MARIPOSA 100

### Compared policies (everything else fixed)

|                 |                                                         |
|-----------------|---------------------------------------------------------|
| <b>cEI</b>      | Myopic expected improvement × posterior feasibility.    |
| <b>cKG</b>      | One-step expected gain in the gated best efficacy.      |
| <b>cEI-tMSE</b> | Alternates gated cEI with gated tMSE boundary sampling. |

## 3. Why lookahead concentrates at a binding boundary

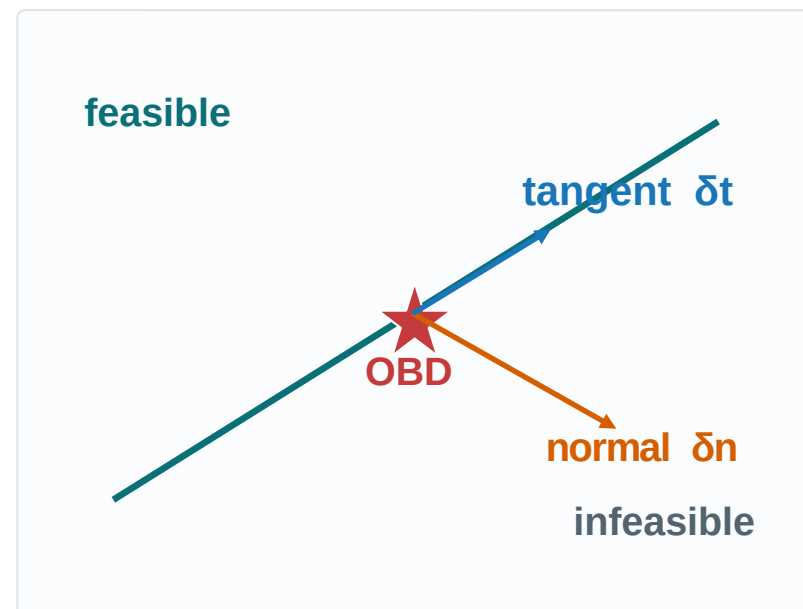
At a boundary dose, the cEI feasibility weight is  $\sigma_g$ -agnostic; strict-gate cKG is  $\sigma_g$ -aware.

$$\frac{\partial \alpha_{\text{cEI}}}{\partial \sigma_g} = 0 \quad (\mu_g = g^\dagger) \quad \text{At the boundary, cEI's feasibility factor is exactly 1/2.}$$

$$\frac{\alpha_{\text{cKG}}(d)}{h - b} = \Phi\left[-\Phi^{-1}(\tau) \frac{\sigma_{y_g}}{\sigma_g}\right], \quad \tau > \frac{1}{2}$$

Local reduction:  $h$  = candidate efficacy;  $b$  = feasible incumbent.  $\sigma_g$  = posterior SD;  $\sigma_{y_g}$  = observation-noise SD. Vol increases with  $\sigma_g$ .

### Constraint-binding geometry at the OBD ( $\lambda$ : shadow price; $H_{\text{tan}}$ : tangential curvature)



$$V(d^* + \delta_n) - V(d^*) = -\lambda \|\nabla g\| |\delta_n| + O(\delta_n^2)$$

Across the boundary: first-order value loss

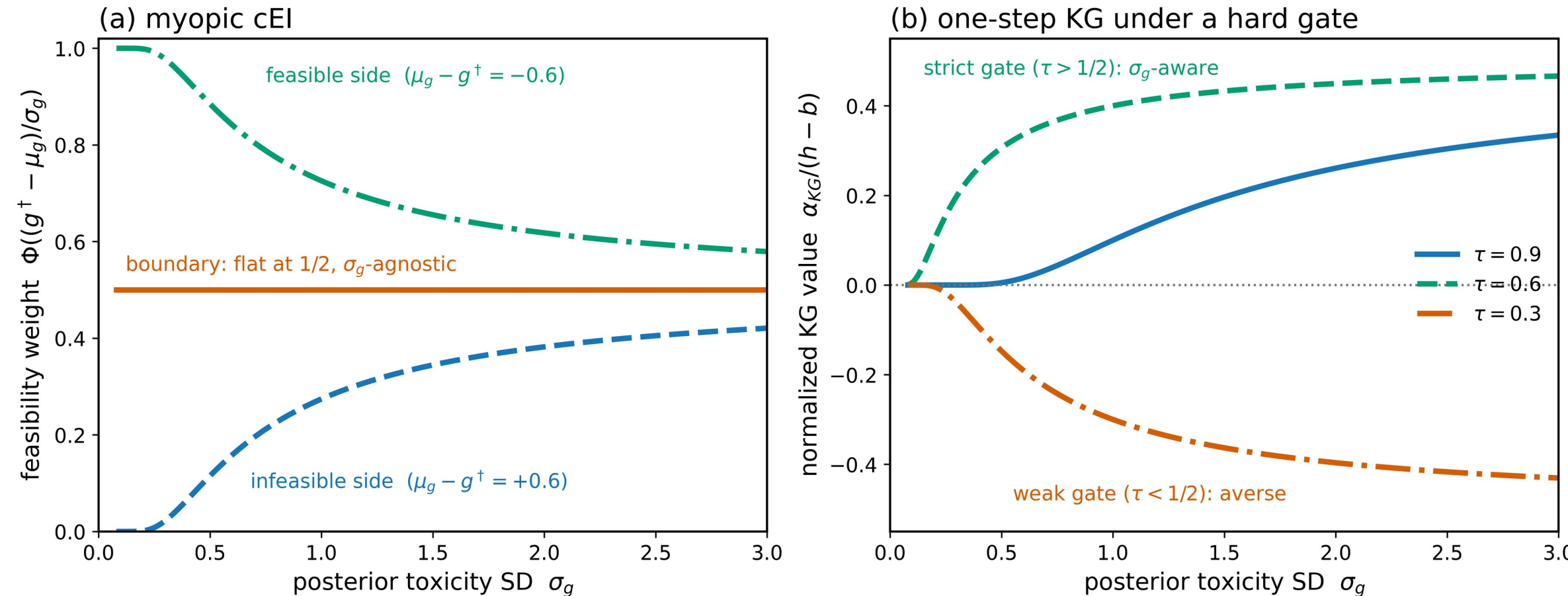
$$V(d^* + \delta_t) - V(d^*) = -\frac{1}{2} H_{\text{tan}} \delta_t^2 + o(\delta_t^2)$$

Along the boundary: second-order value loss

A value-greedy lookahead rule is pulled toward across-boundary uncertainty. It can improve recommendation value without minimizing dose distance or recommendation risk.

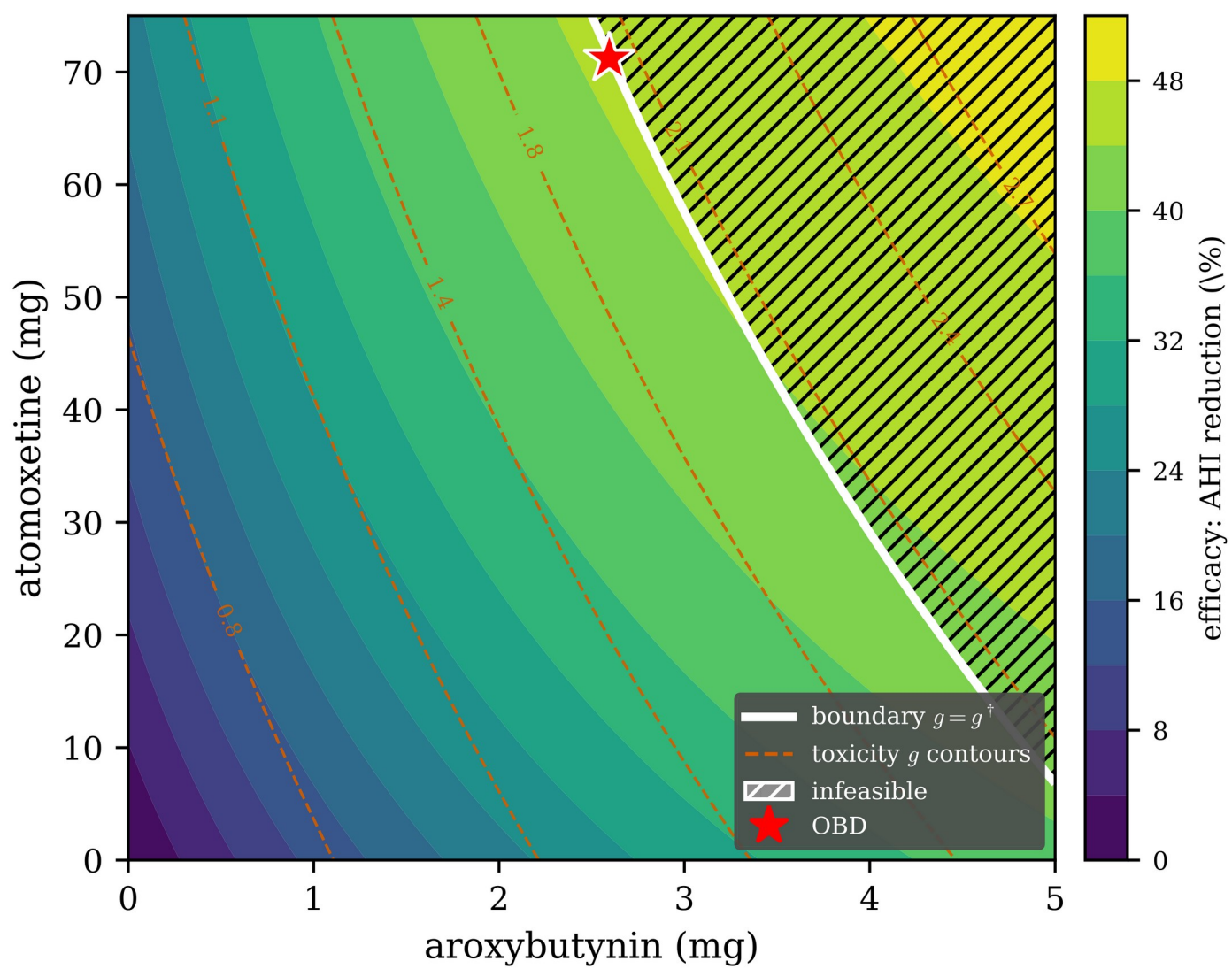
Scope: a directional mechanism at a regular binding optimum - not a claim of universal finite-sample dominance.

## 4. Boundary mechanism (local reduction; adapted from paper Fig. 1)



cEI has no first-order incentive to sharpen boundary uncertainty; cKG can value admitting a high-efficacy boundary dose.  $\tau = 0.3$  is illustrative only (outside the studied gate range).

## 5. Clinically anchored surface: MARIPOSA-informed calibration



Original paper Fig. 2

### Clinical anchor

- Efficacy: percent AHI reduction.
- Toxicity: antimuscarinic adverse-event burden increasing with aroxybutynin.
- Two-drug optimum lies on the tolerability boundary.

### Calibrated OBD

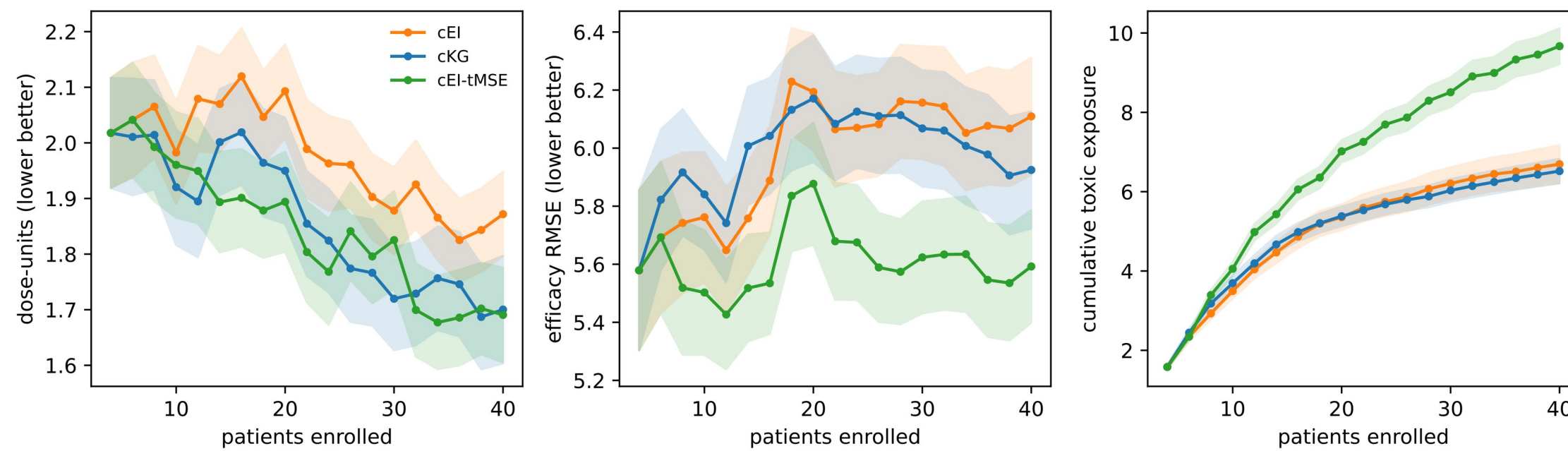
**2.6 mg aroxybutynin**  
**71 mg atomoxetine**  
**45% AHI reduction**

Matches by construction the phase 2b MARIPOSA-selected 2.5/75 mg dose.

### Interpretation boundary

Aggregate calibration only. No patient-level data. This is a clinically anchored consistency check - not independent clinical validation.

## 6. What happens during enrollment? OSA, $\tau = 0.7$



Original paper Fig. 3: 80 paired seed labels pooled over two strata; bands are  $\pm 1$  seed-clustered Monte Carlo SE.

At N = 40 (cEI / cKG / cEI-tMSE): dose-units =  $1.87 \pm 0.08$  /  $1.70 \pm 0.10$  /  $1.69 \pm 0.09$ ; assignments at truly toxic doses =  $6.69 \pm 0.48$  /  $6.52 \pm 0.31$  /  $9.67 \pm 0.45$ .

## 7. Main operating characteristics at $\tau = 0.7$

Means shown; SEs omitted for space. Lower is better except gate precision and efficacy gain.

| Metric / surface                        | cEI         | cKG         | cEI-tMSE    |
|-----------------------------------------|-------------|-------------|-------------|
| OSA unsafe recommendation % (z=0 / z=1) | 13.5 / 6.0  | 23.0 / 21.0 | 7.0 / 6.5   |
| OSA gate precision % (z=0 / z=1)        | 76.2 / 82.1 | 87.8 / 90.8 | 96.2 / 95.2 |
| Logistic dose-units / unsafe %          | 1.18 / 10.5 | 1.32 / 22.5 | 1.15 / 10.5 |
| MARIPOSA dose-units / unsafe %          | 1.90 / 0.0  | 1.54 / 21.0 | 1.63 / 10.0 |

### Multiplicity-aware location result

- Continuous target: 3/6 OSA cEI-cKG contrasts were Holm-significant in the prespecified 12-test family. Post hoc source-panel re-score: 1/6.
- Against cEI-tMSE, cKG had no general localization advantage. On logistic, its mean dose-units were larger at every gate; 3/6 contrasts remained Holm-significant.

### Recommended-dose efficacy gain: cKG minus cEI

|                                                        |                                                             |                                                                      |
|--------------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------------------|
| OSA<br><b>+0.872 ± 0.223</b><br>surface efficacy units | Logistic<br><b>+0.029 ± 0.006</b><br>surface efficacy units | MARIPOSA<br><b>+4.349 ± 0.716</b><br>AHI-reduction percentage points |
|--------------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------------------|

## 8. Objective-specific decision guide for pharma

|                                   |                           |                                                                                          |
|-----------------------------------|---------------------------|------------------------------------------------------------------------------------------|
| Highest recommended-dose efficacy | cKG                       | Highest mean true efficacy at $\tau = 0.7$ on all four surfaces (three shown above).     |
| Final recommendation safety       | cEI or cEI-tMSE           | Avoids cKG boundary aggression; which lookahead-free rule is safer varies by surface.    |
| OBD localization                  | Simulate cEI and cEI-tMSE | No general cKG advantage over cEI-tMSE; logistic results favor lookahead-free rules.     |
| In-trial exposure                 | Evaluate directly         | Exposure ordering differs from final recommendation safety.                              |
| Stricter gate                     | Any candidate policy      | Reduces cKG unsafe recommendations, but may not minimize exposure or localization error. |

## 9. Clinical translation: what is still missing

|                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Current evidence</b> <ul style="list-style-type: none"><li>Acquisition-policy operating characteristics - not a complete trial design.</li><li>Continuous latent outcomes; fixed N; posterior chance gate; no hard elimination/stopping.</li><li>Aggregate clinical anchor; no patient-level validation.</li></ul> | <b>Before deployment</b> <ul style="list-style-type: none"><li>Overdose cap, dose elimination/stopping, and a no-dose option.</li><li>Binary/delayed endpoints and late-onset toxicity handling.</li><li>Calibrated GP uncertainty; prior sensitivity and data-conflict safeguards.</li><li>Clinical utility elicitation, regulatory error metrics, and validated software.</li></ul> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 10. Take-home message

Choose and validate the acquisition for the prioritized clinical objective. Report recommended-dose efficacy, localization, final recommendation safety, and in-trial exposure separately.